The CAPM

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Overview

- The “fundamental value”, “fair value”, of asset is given by

$$FV_t = \frac{E[CF_{t+1}]}{1+r} + \frac{E[CF_{t+2}]}{(1+r)^2} + \frac{E[CF_{t+3}]}{(1+r)^3} + \ldots$$

  - use all available information to make the best possible cash-flow forecast
  - use a discount rate r that is appropriate for the risk

The discount rate

- This lecture note is about the discount rate, r

$$\mathrm{E}(r) = riskless\ rate + risk\ premium$$

- The CAPM measures the risk of an asset
  - provides the fair discount rate r for that risk

- Other applications include
  - Capital budgeting
  - Evaluating investment performance

Why do we hold the market portfolio?

- Suppose:
  - Two stage model
  - Everyone knows the same set of information. All have the same estimates of expected returns, variances, covariances, etc
  - Everyone has the same preference. All use risk-reward analysis (mean-variance analysis)

mean − α variance

"repetitive investors" 相当于只有一人.

他会买所有stock (clear the market) ⇒ need to adjust return.

Why is covariance the right measure?

- Suppose you are holding a portfolio with return r_p
- Add a small amount δ of a new security with return r_i, how does this affect the riskiness of your portfolio?

$$var\big((1-\delta)r_p + \delta r_i\big)$$
$$= (1-\delta)^2 var\big(r_p\big) + \delta^2 var(r_i) + 2\delta(1-\delta)cov\big(r_p, r_i\big)$$
$$\approx var\big(r_p\big) + 2\delta cov\big(r_p, r_i\big)$$

small enough.

- We should measure the risk of a security by how much its inclusion in your portfolio affects your *portfolio's* risk
- A security's risk is therefore driven by its *covariance* with your existing portfolio

The CAPM argument

- A security's risk is its covariance with the rest of the portfolio
  - Variance is a good measure of overall *portfolio* risk
  - For an individual security, *covariance* is the right measure of risk
- The rest of your portfolio is the market portfolio (under some assumptions)
- Why?

The CAPM argument

- A security's risk is its covariance with the rest of the portfolio
  - Variance is a good measure of overall *portfolio* risk
  - For an individual security, *covariance* is the right measure of risk
- The rest of your portfolio is the market portfolio (under some assumptions)
- Risk is covariance with the market portfolio
- **A security's fair average return is determined by its covariance with the market**

Summary of the CAPM argument

- Finance theory says a security's fair average return is determined by its risk
  - We determine a security's "risk" as measured by its covariance with our portfolio
  - "our portfolio" is the market portfolio
  - A security's fair average return is determined by its covariance with the market portfolio.

More details on the CAPM argument

- A security's fair average return is determined by its covariance with the market
  - This relationship is linear, that is for any asset, , we have

$$E[r_i] - r_f = a + b\, cov(r_i, r_m)$$

- To find *a* and *b,* let's apply this asset i to:
  - The risk-free asset, which gives a=0
  - The market portfolio itself

$$E[r_m] - r_f = bcov(r_m, r_m) = bvar(r_m)$$

$$b = \frac{E[r_m] - r_f}{var(r_m)}$$

More details on the CAPM argument

- Putting this all together,

$$E[r_i] - r_f = \frac{E[r_m] - r_f}{var(r_m)} cov(r_i, r_m)$$

- Written slightly differently

risk is not about itself.
But the added risk → covariance

$$E[r_i] = r_f + \beta_i \left(E[r_m] - r_f\right)$$

where $\beta_i = cov(r_i, r_m)/var(r_m)$

We form a portfolio of asset i and the market portfolio M; $(\alpha, 1-\alpha)$, with $\alpha \in [0,1]$. The rate of return is thus $r(\alpha) = \alpha r_i + (1-\alpha) r_M$. Asset i is assumed not efficient; it's point lies in the feasible region but not on the efficient frontier. Thus as α varies this portfolio's point traces out a smooth curve in the feasible $(\sigma, \overline{r})$ region $\alpha \to (\sigma(\alpha), \overline{r}(\alpha))$ parametrized by α, where

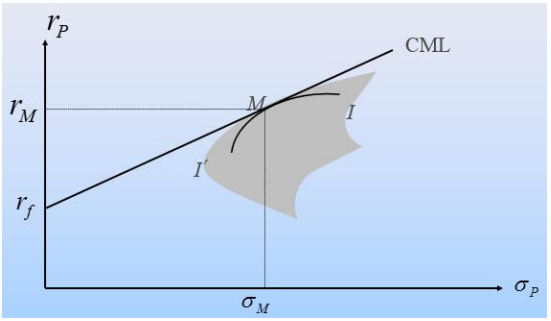
$$\begin{aligned} \overline{r}(\alpha) &= \alpha \overline{r}_i + (1-\alpha)\overline{r}_M \\ &= \alpha(\overline{r}_i - \overline{r}_M) + \overline{r}_M, \qquad (3) \\ \sigma(\alpha) &= \sqrt{\alpha^2 \sigma_i^2 + (1-\alpha)^2 \sigma_M^2 + 2\alpha(1-\alpha)\sigma_{M,i}}. \\ &= \sqrt{\alpha^2(\sigma_i^2 + \sigma_M^2 - 2\sigma_{M,i}) + 2\alpha(\sigma_{M,i} - \sigma_M^2) + \sigma_M^2}. \qquad (4) \end{aligned}$$

When $\alpha = 0$, $(\sigma(0), \overline{r}(0)) = (\sigma_M, \overline{r}_M)$ and when $\alpha = 1$, $(\sigma(0), \overline{r}(0)) = (\sigma_i, \overline{r}_i)$. Thus the curve touches the captital market line at the market point $(\sigma_M, \overline{r}_M)$, but otherwise remains off the captital market line but (of course) within the feasible region where it also hits the point $(\sigma_i, \overline{r}_i)$. We conclude that the curve is tangent to the capital asset line at point $(\sigma_M, \overline{r}_M)$ and therefore when $\alpha = 0$ the curve's derivative

$$\frac{d\overline{r}(\alpha)}{d\sigma(\alpha)}_{\alpha=0}$$

is identical to the slope of the capital asset line at point M. The slope (the price of risk) of the line is given by (2). We thus conclude that when $\alpha = 0$

$$\frac{d\overline{r}(\alpha)}{d\sigma(\alpha)} = \frac{\overline{r}_M - r_f}{\sigma_M}.$$



The composition rule from calculus yields

$$\frac{d\overline{r}(\alpha)}{d\sigma(\alpha)} = \frac{d\overline{r}(\alpha)/d\alpha}{d\sigma(\alpha)/d\alpha}$$

Differentiating (3) and (4) and evaluating at $\alpha = 0$ yields

$$\frac{d\overline{r}(\alpha)/d\alpha}{d\sigma(\alpha)/d\alpha} = \frac{\overline{r}_i - \overline{r}_M}{(\sigma_{M,i} - \sigma_M^2)/\sigma_M}. \quad \Rightarrow \quad \frac{\overline{r}_i - \overline{r}_M}{(\sigma_{M,i} - \sigma_M^2)/\sigma_M} = \frac{\overline{r}_M - r_f}{\sigma_M}$$

The CAPM Equation

- The CAPM says: for any asset, i, the expected return of that asset is

$$\boldsymbol{E[r_i] - r_f = \beta_i (E[r_m] - r_f)}$$

  - r_f is the riskless rate
  - $\beta_i = cov(r_i, r_m)/var(r_m)$ is our measure of risk
  - $\mathrm{E}[r_m] - r_f$ is the *market risk premium* or *equity premium*

The CAPM and discount rate

- The expected return (aka discount rate, required return, fair average return) is:

$$\mathrm{E}[r] = \text{riskless rate} + \text{risk premium}$$

- The CAPM formalizes this:

Expected return — Risk premium

$$\mathrm{E}[r_i] = r_f + \beta_i\left(\mathrm{E}[r_m] - r_f\right)$$

Riskless rate — (Amount of risk) × (Price of risk)

CAPM Intuition

- A high beta stock is riskier, and must therefore offer higher returns to compensate for risk. Why?

investor hate it, pay lower price, higher return.

$\beta_j = \frac{Cov(\tilde{r}_i, \tilde{r}_m)}{Var(\tilde{r}_m)}$ if $\beta_j < 0$ (eg. insurance.)

∵ money in bad state is very valuable.

- Because by moving in the same direction as the rest of your portfolio, it adds variance to your portfolio

- A stock with a beta of 2 means if the market falls by 10% the stock on average falls by 20%, amplifying your losses during **bad time**

- Beta is also known as "systematic risk"

CAPM Intuition

- Systematic vs Idiosyncratic:
- Suppose an asset with only firm specific uncertainty
  - i.e., zero covariance with the market
  - CAPM says a fair average return is the risk free rate
- Only uncertainty related to the overall market deserves compensation because you can diversify the rest away

Implementing the CAPM Equation

$$\mathrm{E}[r_i] = r_f + \beta_i\left(\mathrm{E}[r_m] - r_f\right)$$

- r_f is the riskless rate (historically $\approx 4.5\%$)
- $\mathrm{E}[r_m] - r_f$ is the equity premium ($\approx 6.5\%$)
- β_i needs to be *estimated*, typically $0 \leq \beta_i \leq 2$, but in principle can be anything
- You estimate β_i and the CAPM tells you what the fair expected return, or discount rate, should be for the asset
- We then use this discount rate to discount future cash flows in the present value formula (next lecture)

Where does beta come from?

- Typically we estimate beta using a regression:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i\left(r_{m,t} - r_{f,t}\right) + \varepsilon_{i,t}$$

- The coefficient β_i is the stocks beta
- We use "excess returns" in the regression
  - Accounts for fact that risk free rate changes over time

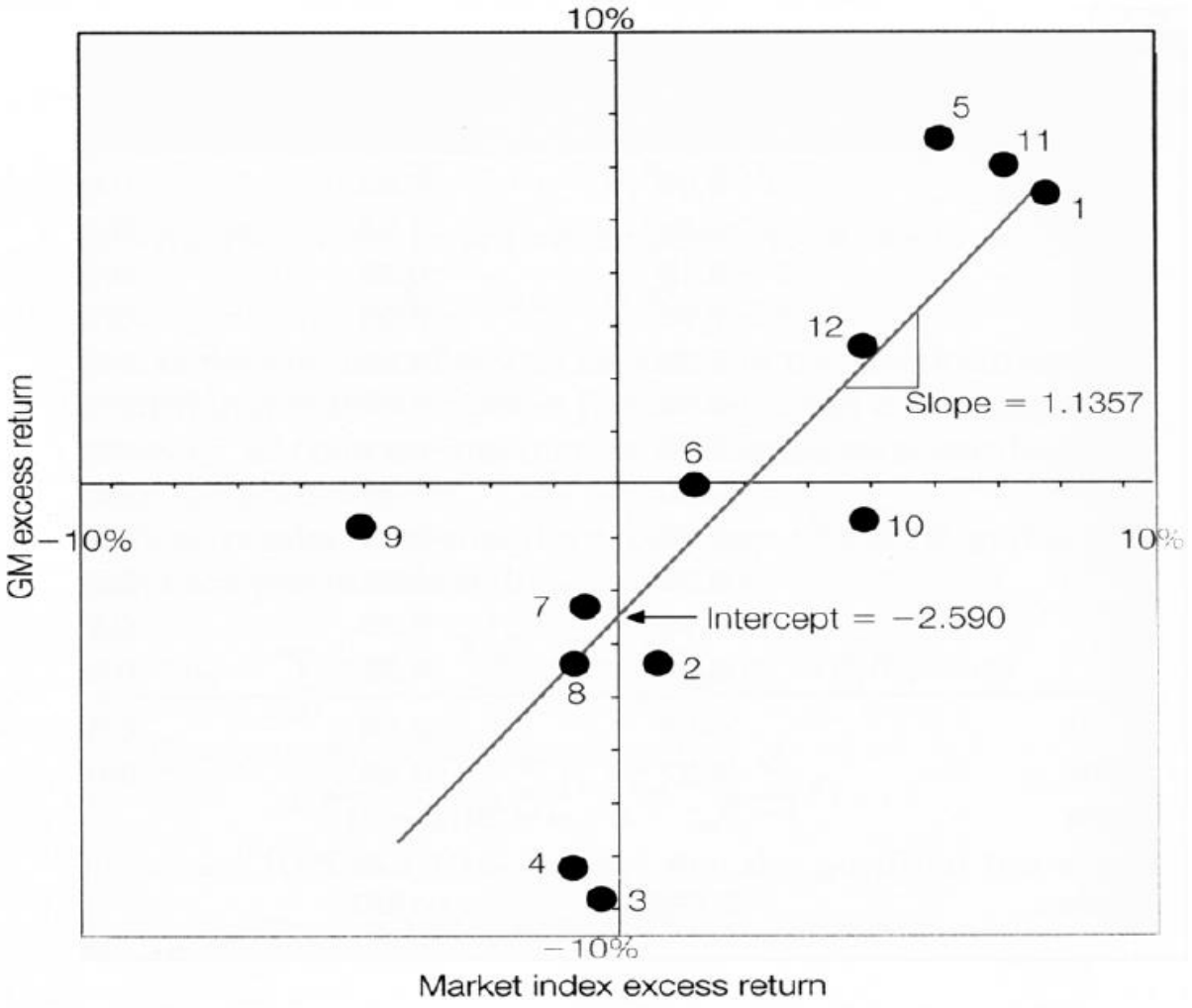
1. use daily data: too much noise
2. use monthly data: sample is small (12 per year) ⇒ use 5 years? ⇒ But firm grows up. β itself already changed.

- Let's do an example with actual data from GM
  - We will use 12 months of data, in practice, it is more common to use 5 years (60 months)

Running the regression

Month	GM Return	Market Return	Monthly T-Bill Rate	Excess GM Return	Excess Market Return
January	6.06	7.89	0.65	5.41	7.24
February	−2.86	1.51	0.58	−3.44	0.93
March	−8.18	0.23	0.62	−8.79	−0.38
April	−7.36	−0.29	0.72	−8.08	−1.01
May	7.76	5.58	0.66	7.10	4.92
June	0.52	1.73	0.55	−0.03	1.18
July	−1.74	−0.21	0.62	−2.36	−0.83
August	−3.00	−0.36	0.55	−3.55	−0.91
September	−0.56	−3.58	0.60	−1.16	−4.18
October	−0.37	4.62	0.65	−1.02	3.97
November	6.93	6.85	0.61	6.32	6.25
December	3.08	4.55	0.65	2.43	3.90
Mean	0.02	2.38	0.62	−0.60	1.75
Standard deviation	4.97	3.33	0.05	4.97	3.32
Regression results	$r_{GM} - r_f = \alpha + \beta(r_M - r_f)$				
	α	β			
Estimated coefficient	−2.590	1.1357			
Standard error of estimate	(1.547)	(0.309)			
Variance of residuals = 12.601					
Standard deviation of residuals = 3.550					
R^2 = .575					

Regression in graphical form



Fair average returns

- Given estimate of beta we can compute the discount rate or fair average return going forward

$$\mathrm{E}[r_i] = r_f + \beta_i\left(\mathrm{E}[r_m] - r_f\right)$$

  - Let's use: r_f=4.5% and $\mathrm{E}[r_m] - r_f$=6.5%, for now.
  - Let's plug in some betas (from Yahoo finance) to calculate fair average returns going forward:

Company	**beta**	**E[r]**
CBRE (real estate)	1.96	17.24%
Moody's (financial)	1.59	14.84%
PetSmart (retail)	0.38	6.97%
Chipotle (restaurant)	0.98	10.87%
Miller (Coors)	1.23	12.50%

What is "alpha" (α)?

- Our regression: $r_{i,t} - r_{f,t} = \alpha_i + \beta_i\left(r_{m,t} - r_{f,t}\right) + \varepsilon_t$
- Take averages of both sides

$$\overline{r_i} - \overline{r_f} = \alpha_i + \beta_i\left(\overline{r_m} - \overline{r_f}\right)$$

  - The error term ε_t is zero on average

- Rearranging,

$$\alpha_i = \overline{r_i} - \left(\overline{r_f} + \beta_i\left(\overline{r_m} - \overline{r_f}\right)\right)$$

Actual average return CAPM implied expected return

$\alpha \neq 0$? reasons:
① may suffer limitation of observation
② investor can make mistake → price wrong
③ reality investors care more factors than μ and σ^2.

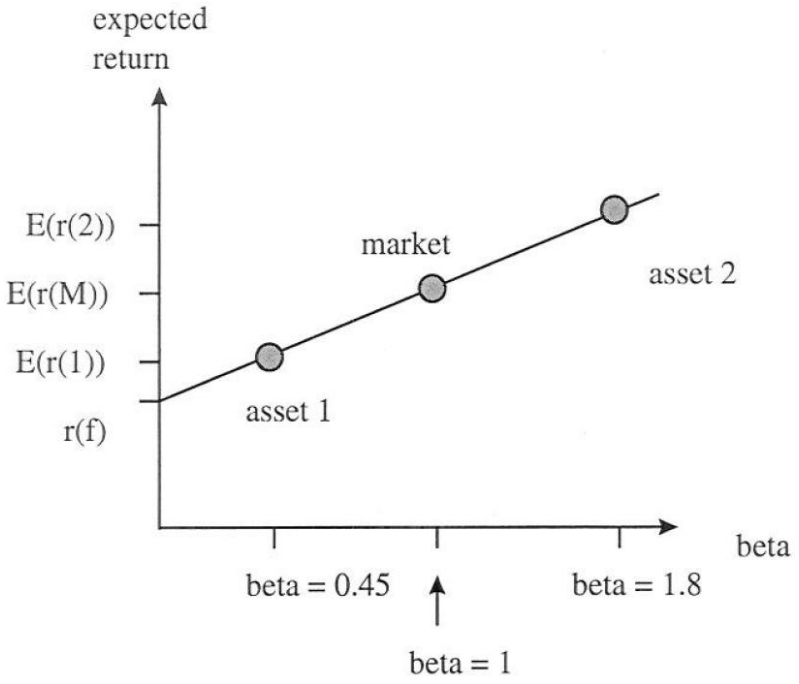
  - $\alpha =$ the *actual* average historical return - the *fair* expected return implied by the CAPM

What is "alpha"?

- The CAPM says **alpha should be zero**
- If a stock or asset has positive alpha (α) it means its average return *historically* has exceeded the fair return implied by the CAPM
- The security may have been mispriced historically
  - Over- or underpriced?
- It may have also just been luck

Does the CAPM work in practice?

- The CAPM says expected returns should line up perfectly with betas



- Let's see if this is true…

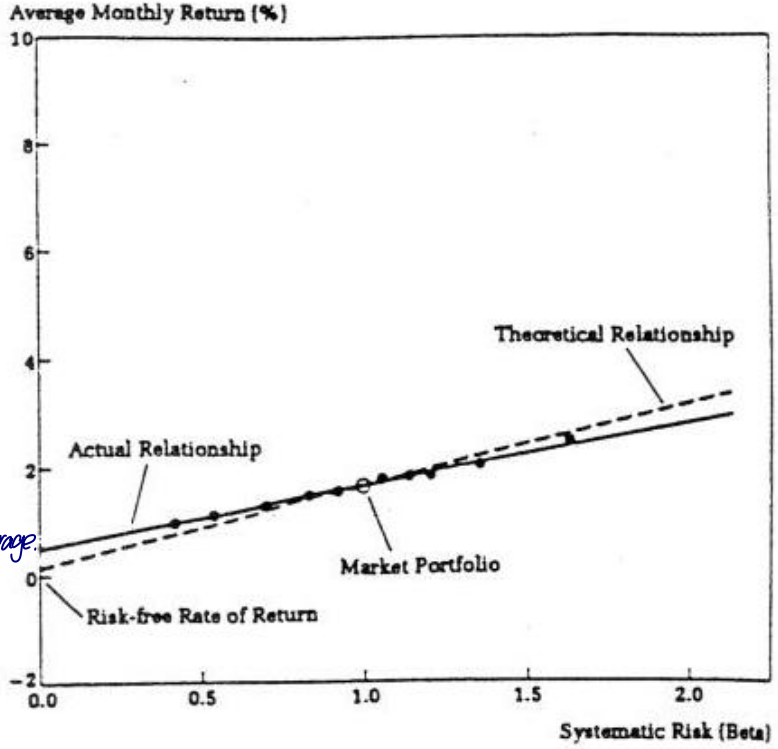
Does the CAPM work in practice?

- Black Jensen Scholes 1972 get mixed results
- The fit is reasonable but not perfect

in CAPM world,
no possibility to generate money from α.
Only way to earn money is to bear more risk.

- want to generate $\beta > 1$? $\Rightarrow$ borrow money (in reality, more complexity).

$\therefore$ investors like (pay higher price) high beta stock. $\because$ higher return, no need to take levarage.



Does the CAPM work in practice?

- Higher beta does correspond to higher return
- But…
  - **low beta** stocks have **higher average returns** than the CAPM predicts
  - High beta stocks have lower average returns than the CAPM predicts

Does the CAPM work in practice?

- Other work by Fama and French 1992
  - Find no relationship between beta and average returns!
  - Declare "**beta is dead**"

- We will use it when we compute fundamental values in the next lecture note

Assumptions of the CAPM

- Individuals
  - Mean-variance optimizers
  - Homogeneous expectations => equivalent to 1 representative investor.
  - All assets are publicly traded
- Markets
  - All assets are publicly held
  - All information is available
  - No taxes
  - No transaction costs

Liquidity and the CAPM

- Liquidity: The ease and speed with which an asset can be sold at fair market value

(first order)

- Illiquidity Premium: Discount from fair market value the seller must accept to obtain a quick sale.
  - Measured partly by bid-asked spread
  - As trading costs are higher, the illiquidity discount will be greater.

The Relationship Between Illiquidity and Average Returns

* Dealer vs Broker
  - car dealer : have inventory
  - house broker: no inventory

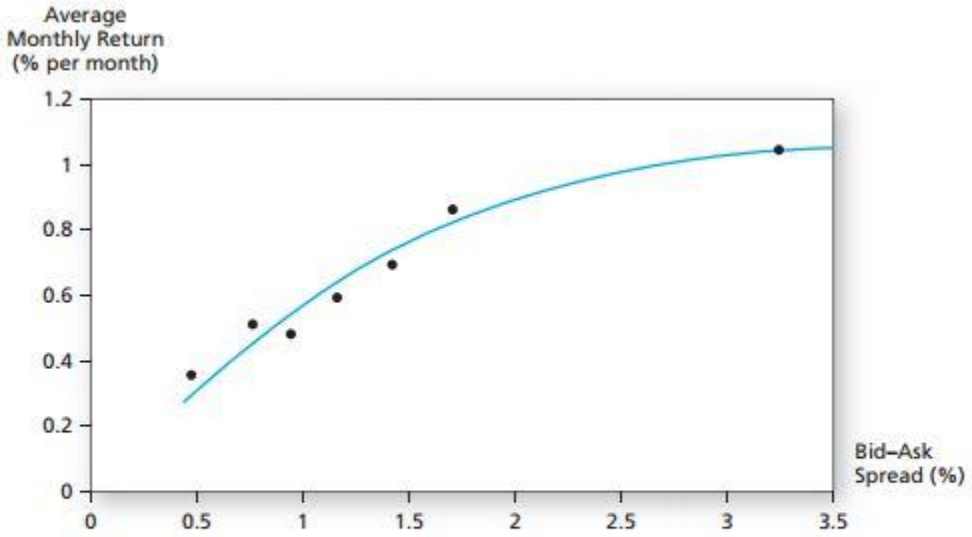


Figure 9.4 The relationship between illiquidity and average returns

Source: Derived from Yakov Amihud and Haim Mendelson, "Asset Pricing and the Bid–Ask Spread," *Journal of Financial Economics* 17 (1986), pp. 223–49. Copyright 1986, with permission from Elsevier.

Liquidity Risk (secondary order).

- In a financial crisis, liquidity can unexpectedly dry up.
- When liquidity in one stock decreases, it tends to decrease in other stocks at the same time.
  - This is systemic
- Investors demand compensation for liquidity risk
  - Liquidity betas

Summary

The CAPM equation: for any asset or stock, i,

$$\mathrm{E}[r_i] = r_f + \beta_i\left(\mathrm{E}[r_m] - r_f\right)$$

- β_i measures risk, *covariance* with the market portfolio
- *Covariance* is what matters because it measures the marginal risk to your portfolio
  - Financial theory says higher risk deserves higher reward
  - The CAPM formalizes this
  - Gives **starting point** for estimating *fair* expected return